Bread and Bicycles

What Happens to an Interaction Claim When the Second Detail Means Nothing

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Abstract

When a person tells a language model two things about themselves rather than one, does the model treat them worse than the sum of the parts? The published literature is split: one camp reports compounding disadvantage at an intersection, a newer camp reports sub-additivity and a threshold past a few attributes. We put the same stacked signals through five outcome measures on three open models (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B), with clause count held constant so that "two signals" never also means "one more sentence".

Alongside every audited pair we stack pairs of socially *neutral* details built the same way, in the same slots: cycling to work and baking bread, walking a dog and learning guitar, making tea and reading history books. Their interaction should be zero. On four of the five instruments it is not. Trait log probability, the rebuilt head-to-head choice and the log probability of the correct answer report significant interactions between two meaningless details, on all three models, on the last of those only within difficulty strata for Qwen, and in places as large as the social ones; the mechanism is visible in the data, in that the effect of a clause depends on the clause beside it. The original head-to-head choice is counted for a different reason: its interaction contrast is invalid by construction, so it yields no interaction result at all, and its floor moves as well. Only one instrument, whether the model solves the person's own arithmetic task, holds a clean zero, on all three models and within difficulty strata.

On that instrument, one interaction of twenty-four survives correction: on Llama, in one slot order, two stated disclosures together cost 7.2 accuracy points *less* than the sum of their separate costs. It is sub-additive, the opposite of compounding, and it is the size of the spread the same measurement shows across six neutral wrappers. Stacking depth produces no shape on any of six ladders. A credential recovers zero to nine percent of the penalty for stating one's age. Asked which of two signals moved them, the models cannot say.

We registered before collection what would confirm and what would refute the hypothesis that signals changing the *text* compound while signals that merely *state* a fact do not. Form-by-form compounding is not detected on any model or instrument that can identify it, which gives no support to half of that hypothesis but falls short of refuting it, since the one clean instrument cannot see small effects; the other half is unreadable on the instruments that fail their floor check.

Four failing instruments could be a fact about our implementation, so we then ran a neutral comparison on published work with the authors' own code, under a second preregistration, reproducing one published number before adding anything. Two targets carry a complete result. In a widely cited study reporting that identity interactions are significant in about a third of regressions, the same statistic returns 0.22 to 0.25 for the adjectives *hungry* and *sleepy*, against a nominal 0.05; two of its four identity groups sit in that range, and two nominally exceed the preregistered neutral-range threshold, by 0.002 and 0.004, against a single shared neutral pair of that size, with no second pair registered; a secondary randomisation test does not separate them

from the neutral pool ($p = 0.33$ and 0.27). A 2025 benchmark reports coreference disparities above 20 percent, and in our rerun of one of its five models a pair of neutral markers reaches 0.350 on type-2 sentences; eleven of twenty attribute-by-sentence-type cells still sit above every neutral set, four of them by less than their own interval width. In that same rerun, on point estimates, its doubly-disadvantaged pattern is largest with no identity marker at all, and adding one mostly reduces it. On four further targets we could not assemble everything a rerun needs, and we report the five artifacts such a rerun turns out to depend on.

The contribution we can defend is methodological: a compounding result from an instrument without a floor check cannot be distinguished from the instrument responding to more text. Four of the five instruments we built do not return zero on that check, and on both published instruments we were able to rerun, neutral details move the statistic as far as many of the social ones. The floor check is released as a standalone tool.

1 Introduction

This is the third paper in a programme about what systems cost the people they were not built around. Paper 1 defined the average-fit loss, the cost of sitting far from the configuration a system was designed for [6]. Paper 2 asked what changes when a language model learns *one* thing about a person, and measured it through seven channels on the same three models and the same one hundred people [7]. This paper asks what happens when there are two.

The question matters because the literature is split, and both camps are plausible. A large body of work reports that disadvantage compounds at an intersection: Guo and Caliskan define an *emergent* intersectional bias as one not predicted from its constituent categories [9], Lalor et al. report amplification at intersections across ten downstream tasks [10], ABLEIST reports up to 58-fold ableist harm amplified for multiply-marginalised candidates [15], and Compounding Disadvantage reports variance rising with each added demographic dimension [14]. A smaller and newer body reports the opposite: Fu et al. state that when two signals are present their effects are "generally sub-additive rather than simply additive" [18], and Kamruzzaman et al. report an over-specification threshold past one to three attributes [19]. Paper 2's accumulation method sat in the second camp with a null.

The literature has not yet directly compared these claims using the same signals, the same models, the same people and the same outcome measures. That is what this paper does, and the adjudication turns out to depend on something prior to the question: whether the instrument can tell two meaningful details from two meaningless ones.

The paper therefore has three parts. Part I puts two stacked signals through five outcome measures on three open models, with a neutral pair stacked the same way beside every audited pair. Four of the five instruments do not meet that criterion. Part II takes up a reasonable objection, that five instruments we built are five instruments we built, and runs a neutral comparison on published work with the authors' own code, under a second preregistration that requires one of their numbers to be reproduced before anything is added. Two targets passed that gate, and in both, neutral details move the statistic well above zero. Part III lists what a rerun of this kind turns out to depend on. On four of the seven targets we could not assemble all of it, and the list is what we now ship with our own work.

What the reader is asked to take away is one sentence, and it is methodological rather than about any particular group: an interaction claim produced by an instrument that has not been shown to return zero for two meaningless details cannot be told apart from the instrument responding to more text.

Terminology. We say *signal stacking* rather than intersectionality. The literature above is about identity categories crossing. Our catalogue is organised by type of deviation and contains items that are not identity categories at all: writing as a non-native speaker, imprecise typing whose cause is unknown, first-time use of a tool. Borrowing a term with a specific social-theory meaning for that set would be wrong, and the distinction earns its place because the variable of interest here is the type of signal, with identity categories as one type among several.

2 Related work, and what this paper concedes

The design is not novel and we do not claim it. Hoffmann et al. apply a full factorial to an LLM hiring audit [33], and Leyva-Vazquez and Smarandache run three factorial studies with bootstrap intervals on interaction magnitudes, documenting a rescue effect in which a prestigious publication compensates for a low-prestige institution [34]. Sub-additivity is not a discovery: Fu et al. published it for identity and opinion [18], and Liu et al. report that two form corruptions of the task text together cost more than either alone but, in most settings, less than the sum [21]. The over specification threshold is published [19]. What is new here is the *contrast* between signal types in one design, the floor check applied to every instrument, and the finding that most of them do not meet the preregistered criterion.

Three prior results shape the methodology. First, instrument validity: Wang et al. show that first-token probabilities mismatch the text an instruction-tuned model actually emits at rates above 60 percent [28], and argue that the text is the more robust read [29]. We answer this rather than assume it: every probability read in this paper records the mass on the answer tokens, the gate is applied in the analysis, and every reported comparison is a difference between two conditions in the same slot, so the slot term cancels. Second, order effects: Yin et al. report that position bias is generally stronger than gender bias and cannot be removed by averaging or randomising presentation [30]. That is the external account of why Paper 2's forced-choice method failed at a position ceiling, and it is why pairs carrying a stated signal run in both slot orders here. Third, cue instability: Tonneau et al. report over 14.8 million prompts that both the magnitude and the direction of a group difference vary with how the group is cued [23]. Our wrapper experiment is the neutral-text version of that warning, and it is the error bar around every other number we report.

The surrounding literature, in one place. The question of how subgroups should be audited at all predates language models: Kearns et al. show that fairness holding on each attribute separately can fail on their intersections, and audit subgroups directly [8], and Gohar and Cheng survey the notions and the mitigations that followed [11]. Within language models the measurements take several shapes: Boufaied et al. study intersectional fairness directly [16], Reddy et al. test it by metamorphic relations [13], and Maltbie and Raval report that false validation of a user's claim varies with the demographics the model perceives [17]. A second group asks what a stated persona does to the model rather than to its verdict on a person: Gupta et al. find implicit reasoning deficits in persona-assigned models [24], and Occhipinti et al. report that sociodemographic prompting helps some groups and hurts others [20]. A third concerns signals that are carried by the text rather than stated, which is our form-against-stated distinction from the other side: Haq and Saldias separate bias from implicit linguistic signals from bias from explicit profiles [22]. A fourth measures what disclosure costs the person who discloses: Kamruzzaman and Kim on disability disclosure in candidate selection [26], Yildirim on mental health disclosure to personalised agents [31], Poole-Dayan et al. on underperformance falling disproportionately on vulnerable users [25], and Panda et al.'s disability benchmark across 21 models [27]. Finally, on the statistics of designs

like ours, Pashley et al. give sample-size determination for 2^K audit experiments with a binary response [32], which is the family our channel A belongs to and the reason its minimum detectable interaction is printed beside every null.

Controls closest to the floor check. Three recent works come closest, and none of them, to our reading, tests an interaction. Vohra and Ravikiran add a placebo to LLM decision audits in which candidates differ only in a decision-irrelevant detail, a hobby or a neighbourhood, and find that under a forced-choice probe the placebo reads as unfair 74 percent of the time, against 5 percent under open-ended scoring [35]; their placebo varies one detail and protects a single contrast. WinoIdentity includes a non-demographic augmentation baseline, adjectives such as "confused" and "relaxed" in the same slots, used for single-axis gender disparity rather than for its doubly-disadvantaged claim [12]. Rennard and Xypolopoulos calibrate intersectional persona effects against sampling noise and against a random unrelated demographic dimension, and find that models keep one to two identity dimensions and discard the rest [36]; their null is demographic rather than neutral. The pattern our answer-probability channel shows, a second clause adding less than the first, has the same shape as the sub-additivity Fu et al. report, who state that "When both signals are present, their effects are generally sub-additive rather than simply additive" [18]; on our instrument that channel does not pass the floor check, which is the point of Section 4. What none of these does, and what the floor check adds, is stack two socially neutral details in the structural positions of the two audited signals and require their interaction to be zero before any social interaction is read. Magee et al. show why it matters: in their data prompt length and the number of terms both correlate with sentiment, and they conclude that intersectional bias "cannot be easily estimated from single-category bias" [37]. Separating those two is what a neutral pair in the same slots is for.

3 Design

Terms. Four words recur and are easy to confuse, so they are fixed here. The *neutral floor* is the interaction estimate for a pair of neutral details; it should be zero. The *floor check* tests whether it is, and an instrument whose floor check rejects is *contaminated*. In Part II, where the published statistic is not an interaction, the *neutral range* is the interval that statistic takes for neutral attributes; it is a comparison band, not a test of zero. The *wrapper spread* is the variation of an estimate across six neutral opening sentences; it measures sensitivity to wording, not contamination.

Models. Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3, loaded at 4-bit on a single free-tier T4. Different developers, different countries, different training data. An effect present in one model only is a fact about that model, and is reported that way.

Signals. Four, drawn from the programme's axes catalogue. Two FORM signals, which change the text the model must read: writing as a non-native speaker (F01) and typing errors (F04). Two STATED signals, which leave the task byte-identical and only assert something about the person: "I use a screen reader for everything on my computer" (S01) and "I am seventy-four and not very confident with computers" (S08). Four 2x2 pairs: form-by-form, stated-by-stated, and two cross pairs.

Clause count is held constant. Every lead-in is exactly two clauses in every condition. A signal takes a slot that a neutral clause would otherwise hold. Without this, "two signals" also means "one more sentence", and the experiment measures verbosity. Paper 2 measured what that costs: a 40-word preamble moved Llama's accuracy by ten points regardless of content.

The interaction. Per item,

$$d = (\text{both} - A) - (B - \text{none}),$$

paired within item. Negative is super-additive, the pair costing more than the sum of its parts, which is the compounding claim. Positive is sub-additive.

The floor. Alongside the audited pairs we stack pairs of socially neutral but distinctive details in the same slots under the same arithmetic: cycling to work with baking bread, walking a dog with learning guitar, making tea with reading history books. Each was checked at build time against a cue-word list; a first draft used "crime novels" and was rewritten because an instrument that reads text reads words. Three floors, not one, because a single neutral pair can itself be unusual.

Instruments. Five, all reading the same conditions.

	what it reads	items	outcome
A	does the model solve the person's task	600 tasks	binary
A2	log P(correct answer) on A's prompts	600 tasks	continuous
B	log prob of six favourable words	100 profiles	continuous
C	head-to-head margin against a twin	200 pairs	continuous
C2	head-to-head, both-cell against each single	200 pairs	continuous

Statistics. Every estimate is a mean with a percentile bootstrap interval and a minimum detectable effect, because an interaction is a difference of differences and carries about twice the variance of a single difference. Each outcome is tested twice: signed rank, and a sign-flip permutation test on the mean, which is the quantity the interval describes. On outcomes with a ceiling the two disagree, and choosing between them after seeing the data would be a forking path, so the rule is fixed in advance and can only remove claims: a floor moves if *either* test rejects, and an effect is claimed only if *both* reject and the mean leans with the signed ranks. Floors are corrected family-wise (Holm) as their own family, one per model and instrument and never together with the contrasts they protect; audited contrasts get Benjamini-Hochberg. The task set was extended from Paper 2's 200 items to 600 after a power analysis measured 20 percent power for the interaction at $n = 200$; rows T001 to T200 are Paper 2's items byte for byte, enforced by a provenance check.

Registered expectations. Written before the first run. *Confirmation:* form-by-form super-additive and clearing the wrapper spread, *and* stated-by-stated indistinguishable from zero, in at least two of three models on at least two of three channels. *Refutation:* stated-by-stated super-additive, or form-by-form absent, in two of three models on channels powered for it. *A null that is still the paper:* no interaction of either type anywhere, with the floor showing that clause count explains what movement there is.

4 The floor result

The three difficulty strata are fixed cuts on each task's mean answer probability across all conditions, at 0.8 and 0.2, and a stratum is used only if it holds at least thirty tasks. The cuts are round probabilities rather than quantiles of these data, but they were chosen after the first A2 analysis, at the same time as the stratified check itself, and the limitations say so. The standalone *Interaction Floor Test* instead splits each floor's items into thirds by the mean of that floor's own four cells, which is a different stratification, and reaches the same verdicts.

Table 1 is the paper's central result.

Table 1: Floor verdicts. Three neutral pairs per instrument per model, corrected within one family per model and instrument: Holm over every floor test, overall and within three difficulty strata, then Bonferroni across the two tests. Twelve families, twelve tests in each of the task channels. The fraction counts the floors that move *in aggregate*; the correction and the verdict also use the stratum tests, which is why Qwen's A2, quiet in aggregate, is marked clean in aggregate only. On A2 a further floor moves within a stratum on Llama and on Mistral without moving in aggregate. Every test behind this table is listed in Appendix A.

model	A accuracy	A2 answer log prob	B traits	C2 choice pairs
qwen	clean	clean in aggregate only	contaminated 3/3	contaminated 6/6
llama	clean	contaminated 2/3	contaminated 3/3	contaminated 6/6
mistral	clean	contaminated 2/3	contaminated 3/3	contaminated 5/5

Channel C is absent from the table because its interaction contrast is invalid by construction: differencing two margins measured against a shared reference does not isolate the extra cost of a second signal. Its floor is large anyway, at +0.50, +1.35 and +3.56 on the three models.

On channel B the floors run from −0.25 to +0.59 and every one of nine is significant. On Llama they do not even share a sign, so the contamination is not an offset that a correction could subtract. For scale, Qwen's neutral floor of +0.553 is larger than its stated-by-stated social interaction of +0.400. On C2 every floor is significant on every model, from +0.35 to +6.44.

Channel A2 fails in a subtler way, and it is the reason the floors are also tested within strata. On Qwen the three floors are quiet in aggregate (+0.18, +0.13, −0.02 against a minimum detectable effect near 0.17), yet floor 1 moves inside the near-ceiling stratum (−0.03, decision p 0.038, Appendix A) while far from the ceiling floors 1 and 2 sit at +0.51 and +0.40. Two movements in opposite directions cancel in the average.

It is not run-to-run noise. The signal-free condition was collected twice, in separate sessions. Channel A gives the same right-or-wrong outcome on 100.0 percent of tasks on all three models; channel B's largest per-profile difference between runs is 0.0053, 0.0024 and 0.0020 log probability, against floor movements of 0.09 to 0.59.

The mechanism. Re-derived independently of the analysers: the effect of a clause depends on the clause beside it. Adding "I cycle to work" to an otherwise neutral message moves Qwen's choice margin by −4.39 logits; adding the same clause to the message that already mentions baking bread moves it by +2.05. On Llama, −0.33 and +0.38. An instrument with that property will report an interaction for any pair whose clauses interact this way, and on these three instruments every neutral pair tested did.

5 What survives: the person's own task

Channel A is the only instrument whose floor is clean on all three models and in every stratum. It is also the least sensitive: at $n = 600$ it detects an interaction of fifteen accuracy points with 90 percent power and one of five points with 13 percent. Every estimate below carries its own minimum detectable effect.

Main effects. Only Llama moves. A stated disclosure costs that model 3.3 to 6.5 accuracy points on the person's own arithmetic (screen reader -0.050 and -0.063 by slot, age -0.065 and -0.033, all $p \leq 0.015$ by exact McNemar), and typing errors cost 2.7 points. On Qwen and Mistral nothing moves, so on those models there is no main effect for an interaction to modify, and the interaction nulls describe the instrument rather than the person's treatment.

Interactions. One of twenty-four survives correction: stated-by-stated on Llama, $+0.072$ accuracy (adjusted $p = 0.004$, minimum detectable effect 0.036), agreeing in sign with the other slot order ($+0.038$, not claimed on its own). It is *sub-additive*: two disclosures together cost seven points less than the sum of their separate costs. Form-by-form is indistinguishable from zero on all three models, with the estimate below the minimum detectable effect in every case.

6 The other instruments, and why they are still reported

Channel B is kept in the paper as a demonstrated failure rather than as a result. It measures a common thing in a common way, the readiness to apply favourable words to a person, and its own floor check rejects on all three models. Its coherence check fails too: on Mistral and Llama several conditions move the favourable words and their opposites in the same direction, which is a scale shift rather than a judgement. Deleting it would leave the methodological argument resting on an assertion about other people's instruments; keeping it makes the argument a measurement.

The main effects of these instruments are unaffected by the floor result, because a single clause replacing a neutral clause has no second clause to interact with. They replicate Paper 2: disclosing a screen reader lowers the readiness to apply favourable words on all three models, by up to two log probability units. Neutral details also lower it, by as much, which is a caution about reading any single-signal penalty against silence rather than against another detail.

7 The rest of the design

Depth (e2). Two ladders, form and stated, at depths 0 to 3, clause count constant. The shape quantity is the second difference $(D_3 - D_2) - (D_2 - D_1)$, bootstrapped over tasks. Six ladders, six intervals covering zero: no shape can be claimed anywhere. Two cells of eighteen move at all, both on Qwen's stated ladder, where two or three signals cost about three accuracy points against none while one costs nothing measurable. That is a threshold rather than a slope, and it sits inside the wrapper spread below.

Wrappers (e1w). Every e1 interaction measured six times, changing only the neutral opening sentence. The spread across wrappers is three to fifteen times the mean, and eight of nine interactions change sign between one neutral opening and another. Nothing clears its own wrapper spread. The one surviving channel A interaction, $+0.072$, is the size of that contrast's wrapper spread, 0.080, which is why it is reported as a direction with a stated limit.

Legibility (e2b). Liu et al. stack two form corruptions of the task text and report a combined drop larger than either factor alone but, in most settings, smaller than the sum, with occasional super-additive cases [21]. Both of their factors corrupt the text the model must read, so we ran the same form-by-form pair in two arms: the signal in the person's message with the task byte-identical, and the styling applied to the task as well. Neither arm is super-additive on any model. On Llama the corrupted-task arm is significantly *sub*-additive ($+0.053$, $p \leq 0.0005$ on both tests), which is the direction Liu et al. report in their own design, and the two arms differ by $+0.063$. With the task byte-identical the interaction is null on all three models. So their sub-additive direction appears on one model of three when the task is corrupted, and nothing appears when only the person's message carries the form signal, with minimum detectable effects of 0.025 to 0.035 stating what this run could have seen.

Offsetting (e3). Does a credential buy back a disclosure? On Llama's three usable numeric measures, stating fifteen years of experience recovers zero to nine percent of the penalty for stating one's age, and in two of six cells the recovery is negative. On the choice channel the answer is slot-dependent: 51 percent recovery on Llama when the credential comes first, nothing on any model when it comes second. The credential is not neutral on its own: on Qwen a candidate who states fifteen years of experience is chosen *less* often than one who says nothing distinctive.

Attribution (e5). Replay the exchange, then ask the model how much each of the two signals changed its answer. Llama answers zero to all 840 ratings: its self-report is a constant and says nothing, the second time a model in this programme has failed that way. Qwen ranks the two signals correctly in 16 of 22 informative items, which does not separate from chance after correction. One of six testable item-level correlations is significant ($\rho = -0.229$). Qwen attributes 9.65 of 10 to cues the person never mentioned, against 3.15 for what the person actually said. Mistral ranks the two signals correctly in 2 of 9 informative items, 22 percent, which does not separate from guessing.

Refusal (e6). Between zero and two refusals per 10,200 generations, at every signal count, on every model. Refusal does not stack on this task.

8 A neutral arm on two rerunnable published instruments

The floor check above is applied to instruments we built. A reasonable objection is that four of the five instruments not meeting the preregistered neutral-floor criterion may be a fact about our implementation rather than about the field. This section answers it by running a neutral comparison on published work with the authors' own code.

What these reruns rest on. Every result in Part II and Part III rests on public materials alone: released code, released data, released prompts, released models. Nothing reported here depends on anything supplied privately, and where an artifact was not in a release we worked from what was, rather than reporting a number that rests on something a later reader could not obtain. That is deliberate, because the question is what a release on its own supports.

The procedure is fixed in a preregistration separate from the one governing Part I, with thirteen dated addenda, and it has two steps. **Step 0, fidelity:** reproduce one published number with the authors' released code and data, under a rule written before the output is read. Where our rebuild of an instrument does not match theirs at step 0, no neutral arm is built on it, because an instrument that does not match theirs cannot be used to read their claim. **Step 1, the neutral arm:** add

attributes that are socially neutral but otherwise built the same way, in the same slots, and read the published claim against them.

Seven targets were selected from thirty-one screened papers. Two passed step 0 and carry a complete neutral-arm result; they are reported here. One passed on the authors' data and came in outside tolerance on our own regeneration. On four we could not assemble everything a rerun needs, and that is Section 9.

8.1 Kirk et al.: what the statistic returns for neutral adjectives

Kirk et al. report that in a regression of predicted occupation on identity terms, "Interactions are significant in approximately one third of regressions" [1]. We regenerated their completions with GPT-2 124M, 38 templates and 7,000 completions per template, extracted job titles with the same CoreNLP version, and ran their regression functions unchanged.

Step 0 passed: the mean share of significant woman-by-religion terms is 0.242 against their 0.203, inside the centred interval of their own released data, $[0.167, 0.245]$.

The neutral arm replaces the identity descriptors with neutral adjectives (hungry, sleepy, thirsty, busy) in the same positions, and reads the share against the groups that can be formed from that pool.

group	terms	social	neutral	neutral interval	verdict
religion	5	0.242	0.247	$[0.199, 0.290]$	inside the neutral range
ethnicity	4	0.224	0.221	$[0.178, 0.270]$	inside the neutral range
sexuality	2	0.289	0.216	$[0.147, 0.287]$	above the neutral range
political	2	0.291	0.216	$[0.147, 0.287]$	above the neutral range

The reading is in the two right-hand columns. For neutral adjectives such as *hungry* and *sleepy* this statistic returns 0.22 to 0.25, where nominal chance is 0.05. For religion and ethnicity the identity share sits in the same range. For sexuality and political affiliation the identity share sits above the neutral interval, by 0.002 and 0.004. **The preregistered rule that decides this is the one stated in the registration: a group counts as above the neutral range when its mean lies above the neutral interval at every matched group size.** Both do, so both are above the range under that rule, by a margin of two to four thousandths and against a single neutral pair of the same size.

A second test was registered as secondary and decides nothing: a randomisation over every size-two group formable from the neutral pool gives $p = 0.33$ and 0.27, at a resolution of 0.067, so it does not separate these two groups from the groups the neutral pool can form. It is reported because it was registered, and it should not be read as support for the result, only as a statement that it cannot distinguish them either way at that resolution.

One possible source of the number is visible in the data. Many per-job regressions have an empty category-by-gender cell: 41 of 64 for the neutral religion group, 41 of 67 for the identity one, and 38 of 63 in the released data. Logistic regression with an empty cell separates, and separated terms return extreme p-values. We read this as a property of the regression rather than of the words in the slot, which would explain why the share is of similar size for the neutral adjectives. That reading is our own, offered as a possible mechanism; the reproduced numbers above stand apart from it and do not depend on it.

8.2 WinoIdentity: where the neutral markers fall

WinoIdentity augments WinoBias referents with identity markers and reports finding "coreference confidence disparities higher than 20% in 7 out of 10 demographic attributes" [12]. We scored one of their five models, Mistral-7B-Instruct-v0.2 at 4-bit, with their prompt format. Step 0 passed: Spearman 0.905 against their 22 published values, mean absolute difference 0.029.

The neutral arm puts neutral markers in the marker slot. Of twenty attribute-by-sentence-type cells, eleven sit above every neutral set and nine are not distinguishable from one. The number to read beside them is the neutral pair itself: *early-rising* and *late-rising* reach a disparity of **0.350** on type 2 sentences, and a three-marker neutral set reaches 0.340. On type 2 sentences, a neutral marker produces a disparity above 20 percent on its own. Four of the eleven cells above the neutral range clear their neutral set by less than the width of their own bootstrap interval.

The second measurement concerns the pattern read as exacerbated bias towards doubly-disadvantaged identities. Writing D for the mean feminine-minus-masculine coreference confidence on male-dominated occupations, more negative meaning the feminine version loses more:

marker	D
no marker at all	$\mathbf{-0.721}$
neutral markers	-0.489 to -0.676
able-bodied	-0.677
non-English-speaking	-0.718
white	-0.264
Asian	-0.241
neurodivergent	-0.245
transgender	-0.179

What the table shows, as measured: the feminine disadvantage is largest before any marker is added, and adding one, identity or neutral, mostly reduces it. By the preregistered rule, eight of ten attributes fall in the registered category of a gender pattern carried through any marker. For the two remaining attributes the most negative marker for disability is *able-bodied*, and non-English-speaking is no more negative than no marker at all.

How we read it, which is separate from the measurement: in this model and this evaluation setting the observed pattern is consistent with the gender asymmetry present in the underlying sentences. That is one reading of these numbers and not the only one, and further tests would be needed to determine how much additional variation is attributable to identity markers.

8.3 What these two reruns do not establish

Both are single-model. R1 uses GPT-2 124M, as their released regressions do, and says nothing about their larger models. R2 uses one of five models at 4-bit. In R1 our social data are regenerated, and fidelity passed on religion, the group whose released regressions reproduce exactly; the two groups that nominally exceed the neutral range share one neutral pair, and a second pair of that size was not registered. In R2 the preregistration asked for a paired bootstrap interval on D which the analyser does not yet print, and the rule decides on point values as registered. Neither rerun shows that a published claim is false, and neither is offered as one. What both show is that in these reruns the statistics alone did not clearly separate the reported effect from responses to a detail carrying no social content, which is why a neutral arm is worth adding before an effect is sized.

9 What a rerun needs

The seven targets were chosen because each released prompts, or data and code. On four of them we could not assemble everything a rerun needs. This section lists what was missing in each case, because it is the cost of Part II and because the same list is what we now ship with our own work.

The claim here is deliberately small. Reproducibility has been measured at scale, on samples of hundreds and thousands of papers, and this is not that. Our seven targets were selected for a different purpose, and seven cases support no rate. What they can offer is a concrete list of the artifacts a rerun of this kind turns out to depend on, which is more useful than a percentage.

9.1 Five artifacts a rerun of this kind depends on

Classifier weights. The LABE metric is computed with three trained classifiers [2]. The released folder carries configuration, scheduler and trainer state for each. The weight files are not part of the public release, so we retrained all three with their own script, reaching validation accuracies of 0.944, 0.933 and 0.941 against the reported 0.933, 0.933 and 0.917. Step 0 then came in at a median absolute relative difference of 0.286 against a registered tolerance of 0.25, and that difference cannot be separated from retraining noise: three classifiers of near-identical accuracy produce variances on the same generations differing by up to a factor of four. The reported ordering, with the intersectional variance largest, appears in 14 of 18 per-classifier rows. That ordering may also be expected from an eight-cell variance containing both the gender and the race spread, which is one reason the neutral arm was designed for this target. With the original weights the question the neutral arm was built for could be settled; without them it stays open.

The input to a preprocessing tool. Wilson and Caliskan code resumes and job descriptions to occupations with a public coding tool [3]. A rerun needs to know which text was sent to that tool, and we could not determine it. We coded 4,859 distinct texts through that tool's API and tried seven readings of the resume side, three registered before the first attempt and four after it failed and before they were computed. None of the seven reproduces the published counts, and the difference has the same shape every time, concentrated in three of the nine occupations. The consistency of that shape is what suggests a single preprocessing choice we did not recover, rather than scattered noise. The job-description side reproduces under the first reading, 8 of 9 counts within 20 percent, so what we were missing is specifically the resume-side input.

A runtime version. Ghanbari Haez and Dragoni generated their answers through a local model server [5]. Which version of it produced the released answers is something we were unable to establish, and generation is sensitive to it. Our regeneration missed the registered fidelity tolerance by 0.0006. A serial rerun, registered as a diagnostic before it was run, did not close the gap: their step is 0.8583, our parallel run 0.8680, our serial run 0.8721. Request batching is therefore not the explanation, and no conclusion about their claim is drawn. What our own generation does show is reported separately and is a statement about our answers only.

How the comparison groups were formed. ABLEIST reports compounding harm across stacked attributes [15]. Its Table 6 caption leaves several groupings open, and all three of our registered readings came in at best 6 of 12 numbers per model against a rule of 10. One mixed reading, which we reconstructed by search over the readings the caption leaves open, reproduces all 24 numbers exactly on both models: the three single-and-double columns are unrestricted, covering every gender and both castes, while the triple-stacked column is restricted to the marginalised values.

Since that reading reproduces every published number to the second decimal on two independent models, it is very likely the computation behind the table, though we cannot establish that from the released materials alone. It also indicates that the composition of the comparison groups may influence the observed increase, and that this is worth considering when interpreting that row. The reconstruction did not satisfy the preregistered reproduction criterion, and the reconstructed interpretation is reported alongside that outcome.

The frame a fine-tuned model expects. The same work releases a fine-tuned labeller. Running it needs the instruction string it was trained with, which we inferred from the released inference code. The labeller could not in the end be run at the scale our check needed, which is a compute limit on our side and is reported as one.

9.2 Beck et al.: how close the reproduction comes

Beck et al. report that combining all sociodemographic attributes produces the largest change in predictions for most of their datasets [4]. Their code, their sampling notebook and six small open models make this the one target that is fully runnable, and we ran their pipeline unchanged over six models and seven conditions.

The registered reading required the printed winner to be ours and our combined share to be within five percentage points, on at least four of six models. Three of the six meet both conditions. On a fourth the combination is also the most influential condition, so the direction of the claim holds on four of the six, but our share for that model is 13.02 against 7.00 and so misses the five-point band. The two remaining models put a single attribute first instead: one changes between 0.0 and 0.1 percent of labels in every condition, so its ordering turns on a single label, and the other is inside the five-point band on size while ordering gender first by 2.1 points.

The preregistration names a fallback reading, the same comparison averaged over all three of their prompt formats, to be used if the first fails. **It is not reported here, because its collection was not finished in time.** The first format is complete and is what the paragraph above rests on; the other two are partial. No number from them appears in this paper, and the fallback reading stays open rather than being decided on part of its data.

What the first format does carry is a check on this work rather than on theirs. The collection was lost to a deleted working directory in October and rebuilt from nothing: a new environment at a new path from the versions the authors pin, the source data downloaded again, the sample rebuilt from their notebook. All thirty-six cells of the recollection reproduce the first to the second decimal, and the verdict is unchanged. The distilled predictions are in the repository with a manifest of checksums, so the table above can be recomputed without the raw collection.

9.3 What this section is for

It is not a reproducibility audit and it is not a judgement on any of these studies, none of which is shown here to be wrong. Of the five artifact types listed above, two could be reconstructed or inferred from what was released, with different levels of confidence and neither fully verified: the grouping behind the published cells, which reproduces every published number but was found by search after the registered readings did not match and which did not meet the preregistered criterion, and the frame a fine-tuned model expects, which we inferred from the released inference code but could not verify at the scale our check needed. The other three could not, which is worth knowing in both directions. The point is forward-looking and applies first to us: a rerun of an interaction claim turns out to depend on a short and predictable list, namely the trained weights

behind any learned metric, the exact input to any preprocessing tool, the runtime version used for generation, the grouping behind each published cell, and the frame a fine-tuned model expects. Part I and Part II of this paper ship each of these that applies to them, and that is the habit we would encourage.

10 Discussion

The registered expectations, answered. Form-by-form compounding is not detected on any model or instrument that can identify an interaction. That provides no support for the half of the hypothesis that says text-changing signals compound, and it is weaker than a refutation: the registration asked for absence on channels powered for it, and the one instrument with a clean floor detects fifteen accuracy points with 90 percent power and five points with 13 percent, so a small compounding effect would not have been seen. The other half, that stated signals do not compound, is not contradicted, but it is not confirmed either. Only one instrument can identify an interaction, it is powered for large effects only, and on one model it finds a stated-by-stated interaction that is non-zero and sub-additive, which does not meet the registered criterion of indistinguishable from zero. On the other four instruments the question cannot be answered. The outcome that matches the data is the third registered one: no compounding is detected on the instruments that pass the floor check, and the floor check indicates that, for several instruments in this study, the observed movement cannot be cleanly separated from responses to additional text.

What the split in the literature might be. Our data are consistent with a simpler account than either camp's. On channel A2, on every model, the interaction takes the sign opposite to the two single effects: neutral clauses cost and two of them cost less than the sum; stated signals raise the answer probability on two models and two of them raise it less than the sum. Of twelve interactions that pass both tests on that channel, seven are a benefit that does not add, two are a cost that does not add, three have singles pointing opposite ways, and none is compounding harm. A second clause adds less than the first, whichever way the clauses push. An instrument that moves for any pair of clauses would produce a pattern of this shape whatever the clauses said, with the sign set by the direction of its single effects. That is a reason to add a neutral arm before sizing an interaction, rather than a reason to doubt any particular result.

None of this is evidence for a general sub-additive effect. The one interaction on a clean instrument is one model, one instrument and one slot-order pair, and it is the size of its own wrapper spread. The A2 pattern is offered as the shape a text-sensitive instrument would produce, not as a finding about how signals combine.

What the published reruns add. A reasonable objection to Part I is that five instruments we built are five instruments we built. Part II answers it in the most direct way available, by running a neutral comparison on other people's published instruments with their own code, after reproducing one of their numbers first. In both targets that passed that gate, neutral details move the statistic well above zero: a statistic whose neutral value is five times nominal chance, and a threshold that a neutral marker reaches. In the second, in our rerun of one of its five models and on point estimates, the pattern read as compounding at an intersection is largest before any identity marker is added. We do not claim either published result is false. In these reruns the statistics alone did not clearly separate the reported effects from responses to socially neutral details. Neutral-condition comparisons may therefore help clarify the interpretation of such results.

What a failed floor check does not show. It does not show that a social interaction is an artifact. It shows that the instrument has a non-zero baseline interaction, which has to be accounted for before the social one can be sized; on channel B even that is not possible, because its floors do not share a sign.

What a reader should take from this. Before reporting that two characteristics compound, stack two socially meaningless details the same way and show that nothing happens. On four of the five instruments here the neutral pair moves the outcome, and on both published instruments we were able to rerun it moves the outcome too. The check costs one extra condition set, and it adds a useful diagnostic for assessing whether an observed interaction reflects the target phenomenon or sensitivity to additional text.

11 Limitations

- Three open models at 4-bit, one task family for the behavioural channel, four signals, two of each type. Nothing here speaks to proprietary models, longer conversations, or signals outside this catalogue.
- Single-wrapper estimates carry the spread measured in e1w as unreported uncertainty. That includes every number in this paper except the wrapper experiment itself.
- Channel A is not powered for small interactions: five accuracy points are invisible at this n, and the paper says so beside every null rather than calling them zero.
- The one surviving interaction is one model, one channel, one slot order pair, and the size of its own wrapper spread.
- The stratified floor check was added after seeing the first A2 analysis. It can only remove claims, and it was validated on synthetic data built to reproduce the failure before being applied, but it was not registered in advance. The standalone *Interaction Floor Test*, which stratifies differently, reaches the same verdicts.
- Mistral's letter-probability reads are heavily gated (32 percent of C2 reads carry less than half the mass on the two letters), so its numbers on that channel rest on as few as twelve items and are reported with their n.
- Part II rests on two reruns, not on a survey. Both are single-model: one uses the 124M model whose regressions its authors released, the other one of five models at 4-bit. Neither says anything about the larger models in either paper, and two cases cannot show how common the problem is.
- In the first rerun the two groups that nominally exceed the neutral range share a single neutral pair, and a second neutral pair of that size was not registered. In the second, the preregistration asked for a paired bootstrap interval on the doubly-disadvantaged statistic which the analyser does not yet print; the rule decides on point values, as registered, and four of the eleven cells above the neutral range clear their neutral set by less than the width of their own bootstrap interval.
- Part III lists five artifacts a rerun of this kind depends on, drawn from seven targets. Seven cases selected for a different purpose support no rate, and nothing there should be read as one.

12 Reproduction

Every number in this paper is produced by scripts in the accompanying repository from the collected CSVs, with no manual steps. The analysis is reproduced by running the analysers in `experiments/*/scripts/` against `experiments/*/outputs/`; the offline validation suite (`validate_*.py`, `reverify_findings.py`, `check_provenance.py`) runs without a GPU and checks the statistics on synthetic data with known answers, re-derives every headline number independently of the analysers, and verifies that the items inherited from Paper 2 are unchanged. The floor check is also released as a standalone tool, *Interaction Floor Test*, which reaches the same verdicts on these data through a different stratification.

Part I and Part II are governed by two separate preregistrations, `PREREGISTRATION.md` and `PREREGISTRATION-PART2.md`, the second with thirteen dated addenda. They are published as written and are not merged retroactively, which is the point of having them: every reading tried is in them, including the ones that failed, and the order in which they were tried was fixed before the data were seen. Each carries the date and the commit hash of the commit that first contained it, in a private working repository that is not published, because its history also holds material that is not ours to publish. Both documents are archived with their own DOI, Part I at 10.5281/zenodo.23959847 and Parts II and III at 10.5281/zenodo.23960058, which makes them immutable from the date of that deposit onward. Neither that deposit nor a commit hash proves the earlier date on which each was written: that much is asserted, supported by a history we can show on request, and the paper claims no more than that. Each target's reruns live in `experiments/r*/`, beside the `e*` experiments of Part I, with the same shape: scripts, outputs, and a result document stating what cannot be claimed.

Acknowledgements

Part II of this paper exists only because seven research groups released their work in a form that someone else could run. We reran instruments built by Kirk et al. [1], by the authors of WinoIdentity [12], by Wan and Chang [2], by Wilson and Caliskan [3], by Ghanbari Haez and Dragoni [5], by the authors of ABLEIST [15], and by Beck et al. [4]. In every case we used their code, their prompts, their data or their released model, unchanged wherever we could. Releasing those materials is what made an independent check possible at all.

Section 9 lists the artifacts a rerun of this kind depends on. It is a list of what is easy to leave out, written by an author who left several of them out of earlier releases of this work, and it is not a comment on the generosity of the releases above.

A Every floor test behind Table 1

Ninety-eight rows, each carrying both tests: each of three neutral pairs on each instrument and model, in aggregate, and on the two task channels within each difficulty stratum as well. One family per model and instrument, Holm within a family for each test and Bonferroni across the two, exactly as Section 4 describes. The two Holm columns are the per-test values *before* the Bonferroni step; the decision is made on the last column, which is twice the smaller of those two, capped at 1, and a floor moves when that value is below 0.05. The rows are written out by `analyse_e1_floors.py` itself, after the correction that produced Table 1, so the two cannot drift apart; the same rows are in `appendix_a_floor_tests.csv` in the repository.

model	ch.	neutral pair	scope	n	interaction	raw p		Holm p		decision
						rank	mean	rank	mean	p
llama	A	cycling, bread	aggregate	600	+0.0100	0.508	0.587	1.000	1.000	1.000
llama	A	dog, guitar	aggregate	600	-0.0133	0.404	0.468	1.000	1.000	1.000
llama	A	tea, history	aggregate	600	+0.0150	0.401	0.461	1.000	1.000	1.000
llama	A	cycling, bread	near ceiling	160	+0.0125	0.727	0.728	1.000	1.000	1.000
llama	A	cycling, bread	middle	98	+0.0000	1.000	1.000	1.000	1.000	1.000
llama	A	cycling, bread	far from ceiling	342	+0.0117	0.433	0.556	1.000	1.000	1.000
llama	A	dog, guitar	near ceiling	177	-0.0056	1.000	1.000	1.000	1.000	1.000
llama	A	tea, history	near ceiling	177	+0.0565	0.025	0.042	0.279	0.460	0.558
llama	A	dog, guitar	middle	96	-0.0104	0.906	1.000	1.000	1.000	1.000
llama	A	tea, history	middle	96	+0.1250	0.159	0.199	1.000	1.000	1.000
llama	A	dog, guitar	far from ceiling	327	-0.0183	0.109	0.107	1.000	1.000	1.000
llama	A	tea, history	far from ceiling	327	-0.0398	0.007	0.011	0.081	0.134	0.161
llama	A2	cycling, bread	aggregate	600	+0.1401	0.009	0.008	0.027	0.041	0.054
llama	A2	dog, guitar	aggregate	600	+0.1353	<0.001	<0.001	<0.001	0.003	<0.001 ⋆
llama	A2	tea, history	aggregate	600	+0.2139	<0.001	<0.001	<0.001	0.001	<0.001 ⋆
llama	A2	cycling, bread	near ceiling	59	+0.0213	<0.001	0.015	<0.001	0.060	<0.001 ⋆
llama	A2	dog, guitar	near ceiling	59	+0.0270	<0.001	0.003	<0.001	0.017	<0.001 ⋆
llama	A2	tea, history	near ceiling	59	+0.0713	<0.001	<0.001	<0.001	0.001	<0.001 ⋆
llama	A2	cycling, bread	middle	123	-0.1567	0.034	0.033	0.067	0.098	0.134
llama	A2	dog, guitar	middle	123	-0.0174	0.080	0.783	0.080	0.783	0.160
llama	A2	tea, history	middle	123	+0.1248	<0.001	0.103	0.001	0.207	0.002 ⋆
llama	A2	cycling, bread	far from ceiling	418	+0.2442	<0.001	<0.001	0.003	0.006	0.005 ⋆
llama	A2	dog, guitar	far from ceiling	418	+0.1955	<0.001	<0.001	<0.001	0.002	<0.001 ⋆
llama	A2	tea, history	far from ceiling	418	+0.2603	<0.001	<0.001	<0.001	0.001	<0.001 ⋆
llama	B	cycling, bread	aggregate	100	-0.0908	<0.001	<0.001	<0.001	<0.001	<0.001 ⋆
llama	B	dog, guitar	aggregate	100	-0.2535	<0.001	<0.001	<0.001	<0.001	<0.001 ⋆
llama	B	tea, history	aggregate	100	+0.2080	<0.001	<0.001	<0.001	<0.001	<0.001 ⋆
llama	C2	cycling, bread	aggregate, vs a	200	+0.6303	<0.001	<0.001	<0.001	<0.001	<0.001 ⋆
llama	C2	cycling, bread	aggregate, vs b	200	+0.7013	<0.001	<0.001	<0.001	<0.001	<0.001 ⋆
llama	C2	dog, guitar	aggregate, vs a	200	+0.7966	<0.001	<0.001	<0.001	<0.001	<0.001 ⋆
llama	C2	dog, guitar	aggregate, vs b	200	+0.4688	<0.001	<0.001	<0.001	<0.001	<0.001 ⋆
llama	C2	tea, history	aggregate, vs a	200	+0.3509	<0.001	<0.001	<0.001	<0.001	<0.001 ⋆
llama	C2	tea, history	aggregate, vs b	200	+0.3822	<0.001	<0.001	<0.001	<0.001	<0.001 ⋆
mistral	A	cycling, bread	aggregate	600	+0.0217	0.221	0.252	1.000	1.000	1.000
mistral	A	dog, guitar	aggregate	600	+0.0033	0.843	0.921	1.000	1.000	1.000
mistral	A	tea, history	aggregate	600	-0.0183	0.300	0.341	1.000	1.000	1.000
mistral	A	cycling, bread	near ceiling	263	+0.0228	0.273	0.362	1.000	1.000	1.000
mistral	A	cycling, bread	middle	98	+0.0510	0.531	0.592	1.000	1.000	1.000
mistral	A	cycling, bread	far from ceiling	239	+0.0084	0.637	0.819	1.000	1.000	1.000
mistral	A	dog, guitar	near ceiling	249	+0.0120	0.491	0.650	1.000	1.000	1.000
mistral	A	tea, history	near ceiling	249	+0.0201	0.225	0.330	1.000	1.000	1.000
mistral	A	dog, guitar	middle	102	+0.0196	0.811	0.893	1.000	1.000	1.000
mistral	A	tea, history	middle	102	-0.0588	0.502	0.568	1.000	1.000	1.000
mistral	A	dog, guitar	far from ceiling	249	-0.0120	0.439	0.607	1.000	1.000	1.000
mistral	A	tea, history	far from ceiling	249	-0.0402	0.012	0.024	0.149	0.286	0.298
mistral	A2	cycling, bread	aggregate	600	+0.1474	0.607	0.129	1.000	0.450	0.900
mistral	A2	dog, guitar	aggregate	600	+0.3698	<0.001	<0.001	<0.001	0.001	<0.001 ⋆
mistral	A2	tea, history	aggregate	600	+0.2782	<0.001	0.011	0.004	0.063	0.008 ⋆
mistral	A2	cycling, bread	near ceiling	52	+0.0162	<0.001	<0.001	<0.001	0.007	0.002 ⋆
mistral	A2	dog, guitar	near ceiling	52	+0.0195	<0.001	0.005	<0.001	0.033	<0.001 ⋆
mistral	A2	tea, history	near ceiling	52	+0.1377	<0.001	<0.001	<0.001	0.001	<0.001 ⋆
mistral	A2	cycling, bread	middle	191	+0.0944	0.316	0.098	1.000	0.450	0.900
mistral	A2	dog, guitar	middle	191	+0.1080	0.714	0.090	1.000	0.450	0.900
mistral	A2	tea, history	middle	191	+0.3261	<0.001	<0.001	0.005	0.002	0.004 ⋆
mistral	A2	cycling, bread	far from ceiling	357	+0.1949	0.371	0.223	1.000	0.450	0.900
mistral	A2	dog, guitar	far from ceiling	357	+0.5608	<0.001	<0.001	<0.001	0.002	<0.001 ⋆

model	ch.	neutral pair	scope	n	interaction	raw p		Holm p		decision
mistral	A2	tea, history	far from ceiling	357	+0.2729	0.075	0.124	0.376	0.450	0.752
mistral	B	cycling, bread	aggregate	100	+0.4278	<0.001	<0.001	<0.001	<0.001	<0.001⋆
mistral	B	dog, guitar	aggregate	100	+0.5859	<0.001	<0.001	<0.001	<0.001	<0.001⋆
mistral	B	tea, history	aggregate	100	+0.2515	<0.001	<0.001	<0.001	<0.001	<0.001⋆
mistral	C2	cycling, bread	aggregate, vs a	15	+2.6417	<0.001	<0.001	<0.001	<0.001	<0.001⋆
mistral	C2	dog, guitar	aggregate, vs a	33	+3.3366	<0.001	<0.001	<0.001	<0.001	<0.001⋆
mistral	C2	dog, guitar	aggregate, vs b	12	+1.2018	<0.001	<0.001	<0.001	<0.001	<0.001⋆
mistral	C2	tea, history	aggregate, vs a	127	+4.0434	<0.001	<0.001	<0.001	<0.001	<0.001⋆
mistral	C2	tea, history	aggregate, vs b	29	+0.7942	<0.001	<0.001	<0.001	<0.001	<0.001⋆
qwen	A	cycling, bread	aggregate	600	+0.0000	1.000	1.000	1.000	1.000	1.000
qwen	A	dog, guitar	aggregate	600	+0.0050	0.750	0.829	1.000	1.000	1.000
qwen	A	tea, history	aggregate	600	-0.0150	0.361	0.425	1.000	1.000	1.000
qwen	A	cycling, bread	near ceiling	304	-0.0164	0.180	0.183	1.000	1.000	1.000
qwen	A	cycling, bread	middle	86	+0.1047	0.160	0.209	1.000	1.000	1.000
qwen	A	cycling, bread	far from ceiling	210	-0.0190	0.317	0.456	1.000	1.000	1.000
qwen	A	dog, guitar	near ceiling	291	-0.0241	0.065	0.063	0.720	0.693	1.000
qwen	A	tea, history	near ceiling	291	+0.0137	0.346	0.469	1.000	1.000	1.000
qwen	A	dog, guitar	middle	89	+0.0562	0.520	0.602	1.000	1.000	1.000
qwen	A	tea, history	middle	89	-0.1798	0.044	0.054	0.529	0.648	1.000
qwen	A	dog, guitar	far from ceiling	220	+0.0227	0.225	0.332	1.000	1.000	1.000
qwen	A	tea, history	far from ceiling	220	+0.0136	0.467	0.634	1.000	1.000	1.000
qwen	A2	cycling, bread	aggregate	600	+0.1818	0.867	0.038	1.000	0.378	0.756
qwen	A2	dog, guitar	aggregate	600	+0.1331	0.483	0.121	1.000	0.848	1.000
qwen	A2	tea, history	aggregate	600	-0.0150	0.731	0.859	1.000	1.000	1.000
qwen	A2	cycling, bread	near ceiling	264	-0.0281	0.130	0.002	1.000	0.019	0.038⋆
qwen	A2	dog, guitar	near ceiling	264	-0.0217	0.908	0.060	1.000	0.478	0.955
qwen	A2	tea, history	near ceiling	264	-0.0173	0.205	0.457	1.000	1.000	1.000
qwen	A2	cycling, bread	middle	86	-0.1237	0.407	0.424	1.000	1.000	1.000
qwen	A2	dog, guitar	middle	86	-0.1576	0.100	0.409	1.000	1.000	1.000
qwen	A2	tea, history	middle	86	-0.0600	0.571	0.772	1.000	1.000	1.000
qwen	A2	cycling, bread	far from ceiling	250	+0.5087	0.060	0.013	0.656	0.140	0.279
qwen	A2	dog, guitar	far from ceiling	250	+0.3965	0.052	0.040	0.618	0.378	0.756
qwen	A2	tea, history	far from ceiling	250	+0.0029	0.633	0.988	1.000	1.000	1.000
qwen	B	cycling, bread	aggregate	100	+0.5533	<0.001	<0.001	<0.001	<0.001	<0.001⋆
qwen	B	dog, guitar	aggregate	100	+0.3794	<0.001	<0.001	<0.001	<0.001	<0.001⋆
qwen	B	tea, history	aggregate	100	+0.4640	<0.001	<0.001	<0.001	<0.001	<0.001⋆
qwen	C2	cycling, bread	aggregate, vs a	200	+0.6469	<0.001	<0.001	<0.001	<0.001	<0.001⋆
qwen	C2	cycling, bread	aggregate, vs b	200	+6.4400	<0.001	<0.001	<0.001	<0.001	<0.001⋆
qwen	C2	dog, guitar	aggregate, vs a	200	+2.6731	<0.001	<0.001	<0.001	<0.001	<0.001⋆
qwen	C2	dog, guitar	aggregate, vs b	200	+5.2081	<0.001	<0.001	<0.001	<0.001	<0.001⋆
qwen	C2	tea, history	aggregate, vs a	200	+4.8869	<0.001	<0.001	<0.001	<0.001	<0.001⋆
qwen	C2	tea, history	aggregate, vs b	200	+5.4903	<0.001	<0.001	<0.001	<0.001	<0.001⋆

A star marks a floor that moves: decision p below 0.05.

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